Formation Control of Nonholonomic Mobile Robots Using Graph Theoretical Methods

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Summary. In this chapter, formation control of mobile robots with nonlinear models is considered. Two controllers are proposed with the aid of the dynamic feedback linearization technique, the time-scaling technique and properties of Laplacian matrix. The proposed controllers ensure the group of mobile robots to move in a desired formation. Existing results in formation control using graph theoretical methods are extended to nonlinear systems of high dimensions. Simulation results show the effectiveness of the proposed controllers.

1 Introduction

Cooperative control of multiple systems has received a lot of attention recently due to its challenging features and application importance. To name a few, multiple vehicles are used in rescue mission, moving a large object, troop hunting, formation control, and cluster of satellites [37, 22, 13, 36, 2, 12, 13, 31, 16, 38, 23]. Different control strategies have been proposed, which include behavior based approach, virtual structure approach, leader following approach, and graph theoretical approach.

Arkin [1] studied cooperation without communication for multiple robots foraging and retrieving objects in a hostile environment. Later, this behavior-based approach was extended to formation control of multiple robots [2]. In the behavioral approach [2, 15, 27, 34, 32, 30], the control action for each robot is defined to be a weighted average of the control associated with each desired behavior. Possible behaviors include collision avoidance, obstacle avoidance, and formation keeping. Using this approach, it is easy to derive control strategies even when vehicles have multiple competing objectives. Generally, it is hard to analyze the behavioral performance analytically.

The virtual structure approach was proposed for formation control of mobile robots by Lewis [19]. It is used to formation control of spacecrafts and multi-satellite systems [29, 16]. In the virtual structure approach, the entire desired formation is treated as a single entity. Desired states for each vehicle in the formation can be specified by place-holders in the virtual structure. Using this approach, it is easy to prescribe the coordinated behavior for the group and the virtual structure can maintain formation during maneuvers. However, the requirement of the formation to act as a virtual structure limits the class of potential applications.

The leader-following approach is another important method [35, 8, 9, 22, 33]. In this approach, some mobile robots are designated as leaders while others as followers. The leaders track desired trajectories, and the followers track desired trajectories with respect to the leaders. The advantage of this approach is its simplicity in that the reference trajectories of the leaders are pre-defined and the internal stability of the formation are guaranteed by individual robot's control laws. A disadvantage is that the whole system fails if the leader fails.

The graph theoretical approach was proposed for cooperative control of multiple linear systems by Fax and Murray [11]. Different control laws were designed with the aid of graph theory [24, 10, 7, 17]. Communication links among systems are described by Laplacian matrices. Each vehicle is treated as a vertex and the communication links between vehicles are treated as edges. Stability of the whole system is guaranteed by stability of each modified individual linear system. However, the methods are limited to linear models of vehicle systems.

In this chapter, we discuss formation control of nonholonomic mobile robots using graph theoretical methods. We propose two formation control approaches to achieve formation stability. In the first approach, the robot's model is transformed to a linear system by dynamic feedback linearization. The controller is then designed based on graph theory. In the second approach, with the aid of the time-scaling technique, a time-varying parameter is introduced in the control law. The two controllers ensure the group of mobile robots to move in a desired formation. While most existing results use linear vehicle models, we discuss cooperative control of nonholonomic mobile robots and design global controllers to achieve formation. We extended earlier results using graph theoretical methods to nonlinear systems with high dimensions.

The rest of the chapter is organized as follows. In Section 2, we introduce terminologies in graph theory which will be used in the chapter. In Section 3, the formation control problem is defined. In Section 4, we design two controllers for the above defined problem. In Section 5, simulation results are presented. Finally, we conclude the chapter in Section 6.

2 Graph Theory

In this section, some terminologies and basic properties used in this chapter are listed. Interested readers please refer to literature [6, 21, 5].

A directed graph $\mathbb{G}$ consists of a set of vertices $\mathbb{V}$ and a set of edges $\mathbb{E} \subset \mathbb{V}^2$, where $e = (\alpha, \beta) \in \mathbb{E}$ and $\alpha \in \mathbb{V}$, $\beta \in \mathbb{V}$. The first element of e is denoted $tail(e)$, and the second of element of e is denoted $head(e)$. For all e, we assume that $tail(e) \neq head(e)$, which means that the graph has no self-loop. We also assume that each element of $\mathbb{E}$ is unique. A graph is called un-directed if $(\beta, \alpha) \in \mathbb{E}$ for any $(\alpha, \beta) \in \mathbb{E}$. The in-degree of a vertex α, denoted $d_i(\alpha)$, is the number of edges with α as its head. The out-degree of a vertex α, denoted $d_o(\alpha)$, is the number of edges with α as its tail. A vertex is called balanced if its in-degree is equal to its out-degree. A graph is called balanced if all of its vertex is balanced. Let $\bar{\mathbb{E}}$ be the set of reverse edges of $\mathbb{G}$ obtained by reversing the order of nodes of all the edges. The mirror of $\mathbb{G}$ denoted by $\hat{\mathbb{G}}$ is an un-directed graph with vertices $\mathbb{V}$ and edges $\mathbb{E} \cup \bar{\mathbb{E}}$.

A path on $\mathbb{G}$ of length N from α_0 to α_N is an ordered set of distinct vertices $\{\alpha_0, \ldots, \alpha_N\}$ such that $(\alpha_{i-1}, \alpha_i) \in \mathbb{E}$ for all $i \in [1, N]$. A graph is called strongly connected if any two different vertices α and β in $\mathbb{V}$ there exists at least one path from α to β. A graph is called disconnected if there exists a disjoint subsets of vertices which cannot be joined by any path. The diameter $\mathbb{D}$ of a strongly connected graph $\mathbb{G}$ is the maximum distance between any two vertices of $\mathbb{G}$. For any vertex β in $\mathbb{V}$, the neighbor of β, denoted by $\mathbb{J}_\beta$, is the set of all vertices α such that $(\alpha, \beta) \in \mathbb{E}$.

Assume that the vertices of $\mathbb{G}$ are enumerated and each is denoted as α_i. The adjacency matrix of a graph, denoted as $G(\mathbb{G})$, is a square matrix of size $|\mathbb{V}|$ and defined by

$$G_{i,j} = \begin{cases} 1, & \text{if } (\alpha_i, \alpha_j) \in \mathbb{E} \\ 0, & \text{otherwise.} \end{cases}$$

The degree matrix of a graph, denoted $D(\mathbb{G})$, is also a square matrix of size $|\mathbb{V}|$ and defined by $D_{i,i} = d_o(\alpha_i)$ and $D_{i,j} = 0 (i \neq j)$. Assume the graph $\mathbb{G}$ is strongly connected, D is nonsingular. The Laplacian matrix of the graph is defined as

$$L = I - D^{-1}G.$$

Some properties of Laplacian matrix are very useful [17].

Property 1. Zero is an eigenvalue of L. The associated eigenvector is $\mathbf{1}$.

Property 2. If $\mathbb{G}$ is strongly connected, the multiplicity of the zero eigenvalue is one.

Property 3. All eigenvalues of L lie in a disk of radius 1 centered at the point $1 + 0j$ in the complex plane.

Property 4. If $\mathbb{G}$ is strongly connected, then each nonzero eigenvalues λ of L satisfies

$$\lambda \geq \frac{1}{\mathbb{D}\sum_{i\in\mathbb{V}} d_{i,i}}.$$

Let $A = \{a_{ij}\}_{m\times n}$ and $B = \{b_{ij}\}_{p\times q}$ are two matrices, the Kronecker product of A and B is defined as

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}.$$

By the definition, the following properties are easily derived.

1. given three matrices A, B, and C,

$$A \otimes (B + C) = A \otimes B + A \otimes C,$$

$$(B + C) \otimes A = B \otimes A + C \otimes A,$$

2. If AC and BD exists, then

$$(A \otimes B)(C \otimes D) = AC \otimes BD.$$

3 Problem Statement

Assume there are m mobile robots moving on the plane. For simplicity, all robots are assumed to have the same structure. Noting the results in literature [4, 20], we assume that there exist suitable transformations such that the kinematics of robot $j(1 \leq j \leq m)$ can be transformed globally or locally into the following chained system

$$\begin{cases} \dot{z}_{1,j}(t) &= u_{1,j}(t), \\ \dot{z}_{2,j}(t) &= z_{3,j}(t)u_{1,j}(t), \\ &\vdots \\ \dot{z}_{n-1,j}(t) &= z_{n,j}(t)u_{1,j}(t), \\ \dot{z}_{n,j}(t) &= u_{2,j}(t), \end{cases} \tag{1}$$

with $(z_{1,j}(t), z_{2,j}(t))$ being the X and Y positions of robot j in the $X - Y$ plane.

Remark 1. For many types of wheeled mobile robots, we can always find global or local transformations such that the kinematics of the robots are transformed into chained form (1) with the first two states being the position of the robot in the plane [20]. See Section 5 for one example.

Let graph $\mathbb{G}$ describe the communication links between the group of mobile robots and L_G be the Laplacian matrix corresponding to the graph $\mathbb{G}$. Each vertex represents a robot and each edge represents a communication link between two robots. For each robot i, $\mathbb{J}_i$ denotes its neighbors. In this chapter, we assume that each robot only knows its own state and the relative positions to its neighbors. That is, in the controller design, $u_i(t)$ is assumed to be a function of $z_i(t)$ and $z_i(t) - z_j(t)$ for each $j \in \mathbb{J}_i$.

Given constant vectors $[h_{j,x}, h_{j,y}] \in \mathbb{R}^2 (1 \leq j \leq m)$ which form a desired form $\mathbb{F}$ in the plane. m robots are said to be in formation if

$$z_{1,j}(t) - h_{j,x} = z_{1,i}(t) - h_{i,x} (i \neq j) \tag{2}$$

$$z_{2,j}(t) - h_{j,y} = z_{2,i}(t) - h_{i,y} (i \neq j). \tag{3}$$

In this chapter, the formation control problem is defined as designing control laws $u_i(t)(1 \leq i \leq m)$ such that m robots are in formation as time tends to infinity, i.e., design control laws $u_i(t)(1 \leq i \leq m)$ such that

$$\lim_{t \to \infty} (z_{1,i}(t) - z_{1,j}(t)) = h_{i,x} - h_{j,x} (i \neq j) \tag{4}$$

$$\lim_{t \to \infty} (z_{2,i}(t) - z_{2,j}(t)) = h_{i,y} - h_{j,y} (i \neq j). \tag{5}$$

Remark 2. In the formation control, (2)-(3) means that points $(z_{1,j}(t), z_{2,j}(t))(1 \leq j \leq m)$ form the same form as $(h_{j,x}, h_{j,y})(1 \leq j \leq m)$. Since $(z_{1,j}(t), z_{2,j}(t))(1 \leq j \leq m)$ are positions of mobile robots in the plane, the m robots form the desired formation $\mathbb{F}$.

In the following, we omit the argument t for state variables, for example, $z_{1,i}(t)$ is simply written as $z_{1,i}$.

4 Controller Design

4.1 Approach I

To design the controller, we first transform (1) into a linear system by the dynamic feedback linearization technique.

Lemma 1. *If $\zeta_{1,j}(t) \neq 0 (\forall t)$, by the dynamic controller*

$$\begin{cases} \dot{\zeta}_{1,j} &= \zeta_{2,j}, \\ &\vdots \\ \dot{\zeta}_{n-3,j} &= \zeta_{n-2,j}, \\ \dot{\zeta}_{n-2,j} &= r_{1,j}, \\ u_{1,j} &= \zeta_{1,j}, \\ u_{2,j} &= (r_{2,j} - g_{n-1,j}(z_{4,j}, \ldots, z_{n,j}, \zeta_{1,j}, \ldots, \zeta_{n-3,j}))/\zeta_{1,j}^{n-2}, \end{cases} \tag{6}$$

where $g_{n-1,j}$ is calculated by following recursion

$$\begin{aligned} g_{2,j} &= z_{3,j}\zeta_{2,j}, \\ g_{k,j} &= (k-1)z_{k+1,j}\zeta_{1,j}^{k-2}\zeta_{2,j} + \sum_{l=3}^{k}\frac{\partial g_{k-1,j}}{\partial z_{l,j}}z_{l+1,j}\zeta_{1,j} \\ &\quad + \sum_{l=1}^{k-1}\frac{\partial g_{k-1,j}}{\partial \zeta_{l,j}}\zeta_{l+1,j}, (k=3,\ldots,n-3), \end{aligned}$$

system (1) is transformed into

$$\dot{q}_j = A_v q_j + B_v r_j \tag{7}$$

where $A_v = I_2 \otimes A_0$, $B_v = I_2 \otimes B_0$, $q_j = [q_{1,j}^T, q_{2,j}^T]^T$, $r_j = [r_{1,j}, r_{2,j}]^T$, and

$$A_0 = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}, B_0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, q_{1,j} = \begin{bmatrix} z_{1,j} \\ \dot{z}_{1,j} \\ \vdots \\ \dfrac{d^{n-2}z_{1,j}}{dt^{n-2}} \end{bmatrix}, q_{2,j} = \begin{bmatrix} z_{2,j} \\ \dot{z}_{2,j} \\ \vdots \\ \dfrac{d^{n-2}z_{2,j}}{dt^{n-2}} \end{bmatrix}.$$

Proof: The result can be obtained by direct calculation. ■

Let

$$q = [q_1^T, \ldots, q_m^T]^T, r = [r_1^T, \ldots, r_m^T]^T,$$
$$A = I_m \otimes A_v, B = I_m \otimes B_v,$$

we have

$$\dot{q} = Aq + Br. \tag{8}$$

Let

$$\begin{aligned} \hbar_{j,1} &= [h_{j,x}, 0, \ldots, 0]^T \in \mathbb{R}^{n-1}, \\ \hbar_{j,2} &= [h_{j,y}, 0, \ldots, 0]^T \in \mathbb{R}^{n-1}, \\ \hbar_j &= [\hbar_{j,1}^T, \hbar_{j,2}^T]^T, \\ \hbar &= [\hbar_1^T, \ldots, \hbar_m^T]^T, \end{aligned}$$

and

$$\rho = L_1(q - \hbar),$$

where $L_1 = L_G \otimes I_{2(n-1)}$, we have the following result.

Lemma 2. *If the communication graph $\mathbb{G}$ is directed and strongly connected and $\rho = 0$, the robots are in formation $\mathbb{F}$.*

Proof: Since $L_1(q-\hbar) = 0$, $q = \hbar + a\mathbf{1}$. Therefore, $z_{1,j} - h_{j,x} = z_{n,j} - h_{j,y} (1 \leq j \leq m)$, which means the robots are in formation. ■

Next, we design r such that ρ converges to zero.

Lemma 3. *For system (8), if the communication graph $\mathbb{G}$ is directed and strongly connected, the control law*

$$r = \Gamma L_1(q - \hbar)$$

makes ρ converge to zero, where $\Gamma = I_{2m} \otimes \Gamma_1$, $\Gamma_1 = [g_1, g_2, \ldots, g_{n-1}]$ is chosen such that the matrix $A_0 + \lambda B_0 \Gamma_1$ is stable (Hurwitz) for each nonzero eigenvalue λ of the communication Laplacian matrix L_G.

Proof: We follow the idea in the proof of proposition 4.3 [17]. First, it is easy to prove that the eigenvalues of $A + B\Gamma L_1$ are those of $A_v + \lambda B_v \Gamma_v$ for λ an eigenvalue of L_G. Next, we consider the following system

$$\begin{bmatrix} \dot{q} \\ \dot{\hbar}_p \end{bmatrix} = \begin{bmatrix} A + BFL_1 & -BFL_1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} q \\ \hbar_p \end{bmatrix} := \Pi \begin{bmatrix} q \\ \hbar_p \end{bmatrix} \tag{9}$$

where $\hbar_p = [h_{1,x}, h_{1,y}, \ldots, h_{m,x}, h_{m,y}]^T$. Define the subspace $\mathbb{S}$ by

$$\mathbb{S} = \left\{ \begin{bmatrix} q \\ \hbar_p \end{bmatrix} : L_1(q - \hbar) = 0 \right\}.$$

For any vector $\sigma \in \mathbb{S}$,

$$\sigma = \begin{bmatrix} \mathbf{1} \otimes a + b \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ b \end{bmatrix}$$

where a and b are vectors. Therefore,

$$\Pi\sigma = \begin{bmatrix} \mathbf{1} \otimes A_0 a \\ 0 \end{bmatrix} \in \mathbb{S},$$

which means $\mathbb{S}$ is Π-invariant and the restriction of the transformation induced by Π on $\mathbb{S}$ is $diag[A_0, 0]$. The eigenvalues of the restriction of the transformation induced by Π on the quotient space $\mathbb{R}^{2mn}/\mathbb{S}$ are those of $A_0 + \lambda B_0 \Gamma_1$ for λ a nonzero eigenvalue of L_G. By the assumption that these eigenvalues have negative real parts, the quotient space is stable. Therefore, for any solution $[q, \hbar_p]^T$ of (9), it tends to the subspace $\mathbb{S}$ as time tends to infinity. From the definition of $\mathbb{S}$, $L_1(x - \hbar)$ tends to zero. ■

In Lemma 3, λ may be several different values. Γ_1 is designed to simultaneously stabilize $A_0 + \lambda B_0 \Gamma_1$ for all non-zero eigenvalues λ. Noting the special structures of A_0 and B_0, we have the following lemma.

Lemma 4. *If the communication graph L_G is directed and strongly connected and $n \leq 4$, there always exists $\Gamma_1 = [g_1, \ldots, g_{n-1}]$ such that $A_0 + \lambda B_0 \Gamma_1$ is stable for each $\lambda (\neq 0)$ of L_G.*

Proof: Noting the special structures of A_0 and B_0, we have the following characteristic polynomial

$$det(\alpha I_{n-1} - A_0 - \lambda B_0 \Gamma_1) = \alpha^{n-1} - \lambda g_1 \alpha^{n-2} - \cdots - \lambda g_{n-2}\alpha - \lambda g_{n-1}.$$

By Kharitonov theorem [3], it can be proved that the characteristic polynomial is stable if Γ_1 is chosen as follows.

$$\begin{array}{ll} g_1 < 0, g_2 < 0 & \text{if } n = 3; \\ g_1 < 0, g_2 < 0, -\dfrac{g_1 g_2}{2\mathbb{D}\sum_{i\in\mathbb{V}} d_{ii}} < g_3 < 0 & \text{if } n = 4. \end{array}$$

■

In Lemma 4, we assume $n \leq 4$ because the dimensions of many wheeled mobiles are less than four. For high dimension systems, we can use Kharitonov theorem [3] to find Γ_1 if it exists.

Theorem 1. *For system (1),if the communication graph $\mathbb{G}$ is directed and strongly connected, controller (6) with*

$$r = \Gamma L_1 (q - \hbar)$$

make (4) and (5) satisfied, where $\Gamma = I_{2m} \otimes \Gamma_1$, $\Gamma_1 = [g_1, g_2, \ldots, g_{n-1}]$ is chosen such that the matrix $A_0 + \lambda B_0 \Gamma_1$ is stable (Hurwitz) for each nonzero eigenvalue λ of the communication Laplacian matrix L_G. Moreover, Γ_1 always exists if $n \leq 4$.

Controller (6) solve the formation control of nonholonomic mobile robots with the aid of graph theory. In the formation control, $u_{1,j}$ is required to be non-zero all the time, which means that each robot is required to move along X-axis. By the proof, this can be guaranteed by suitably choosing the control parameters and the initial conditions of the dynamic controller.

4.2 Approach II

To overcome the singularity of the controller, we propose another controller in two steps under the condition that the communication graph is directed, balanced, and strongly connected. In the first step, $u_{1,j}(1 \leq j \leq m)$ are designed such that (4) is satisfied based on the properties of Laplacian matrix. In the second step, $u_{2,j}(1 \leq j \leq m)$ are designed such that (5) is satisfied with the aid of the time-scaling technique and the properties of Laplacian matrix. In the first step, we have the following lemma.

Lemma 5. *For system (1), if the communication graph is directed and strongly connected, control law*

$$u_{1,j} = \dot{\alpha} - \sum_{i\in\mathbb{J}_j} (z_{1,j} - z_{1,i} - h_{j,x} + h_{i,x}), 1 \leq j \leq m \tag{10}$$

make (4) satisfied, where $\alpha(t)$ is any differentiable function.

Proof: Let

$$y_1 = [y_{1,1}, y_{1,2}, \ldots, y_{1,m}^T$$

where

$$y_{1,j} = z_{1,j} - \alpha - h_{j,x},$$

with the control law (10), we have

$$\dot{y}_{1,j} = -\sum_{i \in \mathbb{J}_j} (y_{1,i} - y_{1,j}).$$

So

$$\dot{y}_1 = -L_G y_1. \tag{11}$$

Since the graph $\mathbb{G}$ is directed and strongly connected, L_G is a symmetric nonnegative definite matrix. There exists an non-singular matrix S such that

$$S^{-1} L_G S = \Lambda = \text{diag}[0, J_2, \ldots, J_m]$$

where J_j is the Jordan form corresponding to eigenvalue $\lambda_j > 0 (2 \leq j \leq m)$. Let the state transformation $\xi = S^T y_1$, we have

$$\dot{\xi} = \Lambda \xi.$$

Therefore,

$$\xi(t) = \begin{bmatrix} \xi_1(0) \\ \xi_2(t), \\ \vdots, \\ \xi_(t) \end{bmatrix}$$

where $\xi_j (2 \leq j \leq m)$ exponentially converge to zero. Noting $y_1 = S\xi$, so

$$y_{1,j}(t) = \sum_{i=1}^{m} s_{j,i} \xi_i(t) = s_{j,1} \xi_1(0) + \sum_{i=2}^{m} s_{j,i} \xi_i(t).$$

Since $L_G S = S\Lambda$,

$$L_G \begin{bmatrix} s_{1,1} \\ s_{2,1} \\ \vdots \\ s_{m,1} \end{bmatrix} = 0$$

Noting L_G is a Laplacian matrix corresponding to a directed strongly connected matrix,

$$s_{1,1} = s_{2,1} = \cdots = s_{m,1}.$$

So,

$$\lim_{t \to \infty} y_{1,i} = \lim_{t \to \infty} y_{1,j} (i \neq j).$$

Therefore, $\lim_{t \to \infty} (z_{1,i} - z_{1,j}) = h_{i,x} - h_{j,x} (i \neq j)$. ■

In control law (10), α is a control parameter which plays an important role in designing $u_{2,j}$ in the next step with suitable assumption.

In the second step, we first have the following lemma.

Lemma 6. *For system (1), if the communication graph is directed, balanced and strongly connected, with the control law (10), $u_{1,j}(1 \leq j \leq m)$ exponentially converge to $\dot{\alpha}$ with the least rate $\lambda_2(L_G)$ (i.e., the least non-zero eigenvalue of L_G). If $\dot{\alpha}(t) \geq \epsilon > 0$ for $t \geq T_1$, there exists a finite time $T_2(> T_1)$ such that $u_{1,j}(t) \geq \epsilon/2(1 \leq j \leq m)$ for $t \geq T_2$.*

Proof: By the proof of Lemma 5,

$$\lim_{t\to\infty} y_{1,j}(t) = c$$

where c is a constant. Since the communication graph is balanced, $\sum_{j=1}^{m} y_{1,j}/m$ is an invariant quantity of system (11) and

$$c = \frac{1}{m}\sum_{i=1}^{m} y_{1,i}(0).$$

Let

$$e = [e_1, \ldots, e_m]^T = y_1 - c\mathbf{1},$$

then

$$\dot{e} = -L_G e \tag{12}$$

and

$$\sum_{i=1}^{m} e_i = 0.$$

Let

$$V = \frac{1}{2}e^T e$$

differentiate it along (12), we have

$$\dot{V} = -e^T L_G e \leq -\lambda_2(L_{\hat{G}})e^T e = 2\lambda_2(L_{\hat{G}})V(\forall e \neq 0.)$$

Therefore, e exponentially converges to zero with the least rate $\lambda_2(L_{\hat{G}})$. Since

$$u_{1,j} = \dot{z}_{1,d} - \sum_{i\in J_j}(y_{1,j} - y_{1,i}),$$

$u_{1,j}$ exponentially converge to $\dot{z}_{1,d}$. Noting $\dot{z}_{1,d}(t) \geq \epsilon$ for $t \geq T_1$, there exists a finite time t_j such that $u_{1,j}(t) \geq \epsilon/2$ for $t \geq t_j$. Let $T_2 = \max\{t_j, 1 \leq j \leq m\}$, $u_{1,j}(t) \geq \epsilon/2(1 \leq j \leq m)$ for $t \geq T_2$. ■

In Lemma 6, $z_{1,j}$ increase monotonically with time for $t > T_2$, which means

$$z_{1,j} \to \infty \Leftrightarrow t \to \infty.$$

Therefore, we can use the following time-scaling technique. Let

$$z_{i,j}^{(1)} = \frac{dz_{i,j}}{dz_{1,j}}, (2 \le i \le n, 1 \le j \le m),$$
$$v = [v_1, v_2, \ldots, v_m]^T = [u_{2,1}/u_{1,1}, \ldots, u_{2,m}/u_{1,m}]^T$$

we have

$$\overline{z}_j^{(1)} = A_1 \overline{z}_j + B_1 v_j, (1 \le j \le m) \tag{13}$$

where $\overline{z}_j = [z_{2,j}, \ldots, z_{n,j}]^T$,

$$A_1 = \begin{bmatrix} 0\,1\,0 \cdots 0\,0 \\ 0\,0\,1 \cdots 0\,0 \\ \vdots\,\vdots\,\vdots\,\ddots\,\vdots\,\vdots \\ 0\,0\,0 \cdots 0\,1 \\ 0\,0\,0 \cdots 0\,0 \end{bmatrix}, B_1 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

Using Kronecker product, (13) can be written as

$$\overline{z}^{(1)} = A\overline{z} + Bv \tag{14}$$

where $\overline{z} = [\overline{z}_1^T, \ldots, \overline{z}_m^T]^T$, $A = I_m \otimes A_1$, $B = I_m \otimes B_1$, $v = [v_1, \ldots, v_m]^T$, and

$$\overline{z}^{(1)} = [\frac{d\overline{z}_1^T}{dz_{1,1}}, \ldots, \frac{d\overline{z}_m^T}{dz_{1,m}}]^T.$$

Define

$$h_j = [h_{j,y}, 0, \ldots, 0]^T \in \mathbb{R}^{n-1} (1 \le j \le m),$$

and let ω_j be as follows

$$\omega_j = (\overline{z}_j - h_j) - \frac{1}{|\mathbb{J}_j|} \sum_{i \in \mathbb{J}_j} (\overline{z}_i - h_i), (1 \le j \le m),$$

then

$$\omega = [\omega_1^T, \ldots, \omega_m^T]^T = L(\overline{z} - h)$$

where $L = L_G \otimes I_{(n-1)}$ and $h = [h_1^T, h_2^T, \ldots, h_m^T]^T$.

Since (5) is satisfied if ω tends to zero. Therefore, we only need to design v such that ω converges to zero. Let the control law

$$v = \Gamma\omega, \tag{15}$$

where $\Gamma = \mathrm{diag}[\Gamma_1, \ldots, \Gamma_1]$, we have the following result.

Lemma 7. *Assume the communication graph is directed, balanced and strongly connected, let* $\Gamma_1 = [g_1, g_2, \ldots, g_{n-1}]$*, if the matrix* $A_1 + \lambda B_1 \Gamma_1$ *is stable (Hurwitz) for each nonzero eigenvalue* λ *of the Laplacian matrix* L_G*, then the control law (15) makes* ω *converge to zero.*

Proof: Along the proof of Lemma 3, it is omitted here. ■

By Lemma 7, we can obtain the control law for time $t \geq T_2$. Next, we give the controller of system (1) for all time.

Theorem 2. *For system (1), if the communication graph is directed, balanced and strongly connected, the control law*

$$\begin{cases} u_1 = \dot{\alpha}\mathbf{1} - L_G y_1, \\ u_2 = \mathrm{diag}[u_{1,1}, \ldots, u_{1,m}] \Gamma L(\overline{z} - h) \end{cases} \tag{16}$$

makes (4)-(5) satisfied, where $\dot{\alpha} \geq \epsilon > 0$*,* $\Gamma_1 = [g_1, \ldots, g_{n-1}]$ *and such that* $A_1 + \lambda B_1 \Gamma_1$ *is stable (Hurwitz) for each nonzero eigenvalue* λ *of the Laplacian matrix* L_G*.*

Proof: By Lemma 5, (4) is satisfied. For $t \geq T_2$, (5) is satisfied by Lemma 7 if the state is bounded at time T_2. It is only needed to show $\overline{z}$ is bounded during time interval $[0, T_2]$. Noting the structure of system (1), $\overline{z}$ is bounded due to the boundedness of $u_{1,j}$ and $u_{2,j}$. ■

Control law (16) solves the formation problem in this chapter. With this control law, the robots move in formation as time tends to infinite. In the control law, α is a design parameter. Since $\dot{\alpha} > 0$, the robots always move on the plane. It should be noted that in this controller there is no singular point. Most of the existing results on the cooperative control of mobile robots with graph theory are based on linear models. This chapter shows how the graph theory can be applied successfully to the cooperative control of nonholonomic mobile robots.

5 Illustrative Examples

Consider five wheeled mobile robots moving on a plane (Fig. 1). The five robots are the same and have the following kinematics (Fig. 2).

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ \tan\phi/l \\ 0 \end{bmatrix} R v_1 + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} v_2 \tag{17}$$

where $q = [x, y, \theta, \phi]^T$ is the system state, (x, y) represents the Cartesian coordinates of the middle point of the rear wheel axle, θ is the orientation of the robot body with respect to the X-axis, ϕ is the steering angle; l is the distance between the front and rear wheel-axle centers, R is the radius of rear

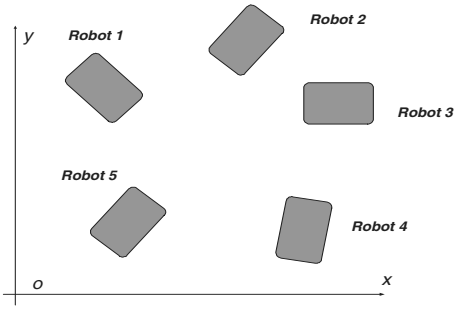


Fig. 1. Five mobile robots on a plane

driving wheel; v_1 is the angular velocity of the driving wheels, and v_2 is the steering velocity of the front wheels. $\phi \in (-\pi/2, \pi/2)$ due to the structure constraint of the robot.

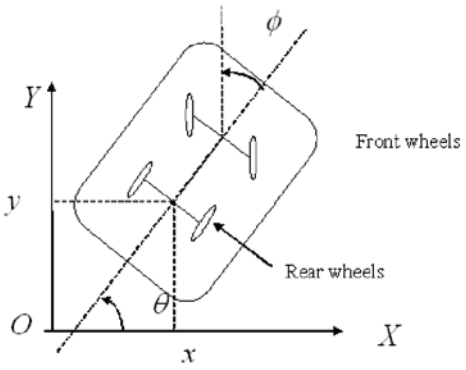


Fig. 2. Structure of a mobile robot

Let the state transformation

$$z_1 = x, z_2 = y, z_3 = \tan\theta, z_4 = \frac{\tan\phi}{l\cos^3\theta}, \tag{18}$$

and the input transformation

$$u_1 = v_1 R\cos\theta, u_2 = \frac{v_2 l\cos^2\theta + 3\sin\theta\sin^2\phi u_1}{l^2\cos^5\theta\cos^2\phi}, \tag{19}$$

system (17) is transformed into

$$\dot{z}_1 = u_1, \dot{z}_2 = z_3 u_1, \dot{z}_3 = z_4 u_1, \dot{z}_4 = u_2. \tag{20}$$

By the dynamic feedback linearization procedure, we can obtained a linear system (7) with $n = 4$. With the results in the last section, we can obtain the controller (6). By the inverse transformation, we can obtain the controller of the original system. In the simulation, the formation is defined by points $(0, 0)$, $(15, 0)$, $(0, 15)$, $(8, 25)$ and $(25, 15)$ (see Fig. 3). Assume the communication link is shown as Fig. 4, Fig. 5 is the simulation result with this information graph. If the communication link is shown as Fig. 6, the group of mobile robots moves as in Fig. 7. The simulation results show that the proposed control law (6) is effective. In the simulation, the singular point never occurs since the robots always move forward.

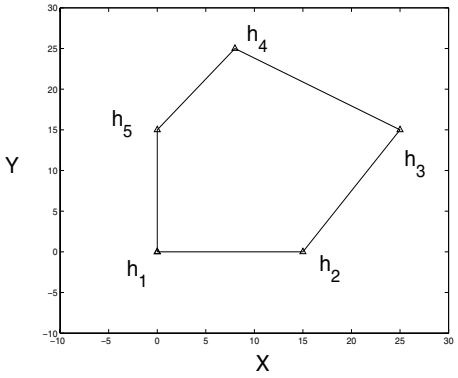


Fig. 3. Desired formation

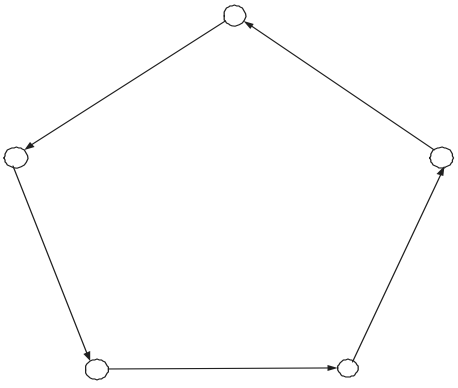


Fig. 4. Information link

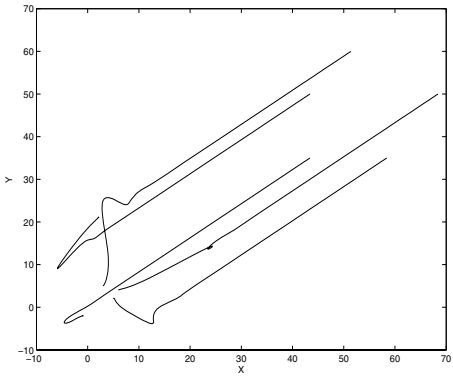


Fig. 5. Trajectories of the group of mobile robots in X-Y plane with controller (6)

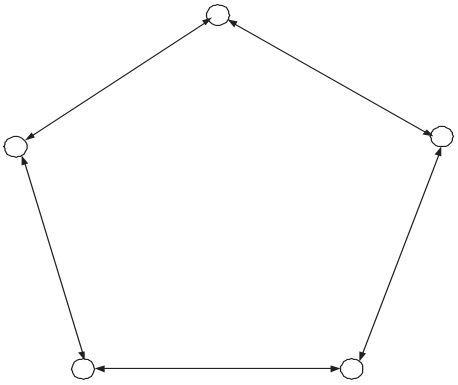


Fig. 6. Information link

We can easily obtain controller (16) too. In the controller, we choose $\alpha = 0.1t$. Fig. 8 is the simulation result with the information graph in Fig. 6. The simulation results show that the proposed control law (16) is effective if the communication graph is connected.

6 Conclusion

This chapter considers the formation control of mobile robots with nonlinear models. Two cooperative controllers are proposed based on the graph theory. Simulation results show the proposed controllers are effective. In this chapter, we do not consider collision and obstacles. How to consider them into controller design is an ongoing research work.

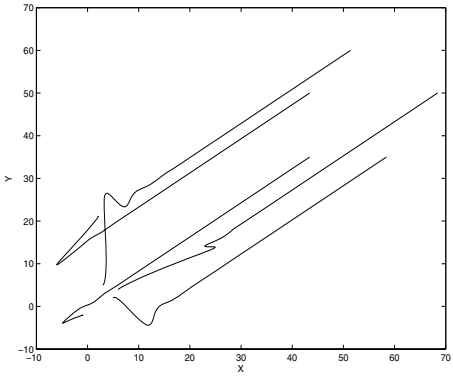


Fig. 7. Trajectories of the group of mobile robots in X-Y plane with controller (6)

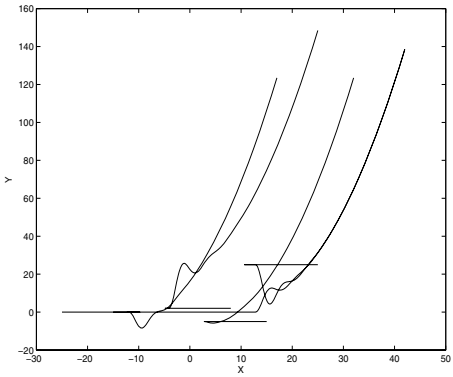


Fig. 8. Trajectories of the group of mobile robots in X-Y plane with controller (16)

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